

ISRM Prior-Art & Standards

Crosswalk

DIN submission support document

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Purpose

This document positions ISRM against major publicly identifiable standards and reference models. It is a scoping aid, not a legal patentability opinion and not a substitute for licensed clause-by-clause standards analysis.

Reference	Primary focus	Overlap risk	Action for ISRM
ISO/IEC 22989:2022	AI concepts/terminology	High terminology relevance	Align definitions; avoid conflicting terms. Public abstract says it establishes AI terminology/concepts.
ISO/IEC 23053:2022	Framework for ML-based AI systems	Medium-high	ISRM broader than ML; map functional concepts.
ISO/IEC 5338:2023	AI lifecycle processes	Medium	Lifecycle vs runtime distinction.
ISO/IEC 42001:2023	AI management systems	Medium	Control Plane can host technical controls; no replacement.
ISO/IEC 23894:2023	AI risk management guidance	Medium	Risk treatment outside core architecture; crosswalk useful.
ISO/IEC/IEEE 42010:2022	Architecture descriptions	High	Use as meta-standard for documenting ISRM viewpoints/models.
NIST AI RMF 1.0	Govern/Map/Measure/Manage	Medium	Govern is explicitly cross-cutting; conceptually supports plane treatment.
NIST 4D/RCS	Hierarchical intelligent control architecture	Very high prior-art relevance	Explicitly acknowledge; differentiate domain neutrality, authority/trust/adaptatio

			n, processing episodes and conformance.
OODA	Observe/Orient/Decide/Act	High conceptual relevance	Different granularity/purpose; OODA not a technical conformance reference model.
BDI	Belief/Desire/Intention agent architecture	High agent-theory relevance	Mental-state/rational-agent model vs operational responsibility model.
MAPE-K	Autonomic computing feedback loop	High conceptual relevance	Monitor/analyze/plan/execute/knowledge overlaps; ISRM adds intent, authorization decision, trust/authority and recursion.
IEEE P3225.05	Internet of Agents reference architecture	Potentially high/future overlap	Active project approved 2026-09-25; coordinate early to avoid duplication.

Novelty claim that should NOT be made

- “First closed-loop model for intelligent systems.”
- “First layered architecture for AI.”
- “First model separating perception, reasoning and action.”
- “First multi-agent reference architecture.”
- “First governance model for AI.”

Defensible contribution statement

ISRM proposes a specific, technology-neutral operational reference model that combines six functional responsibilities (Intent, Perception, Context, Cognition, Decision, Execution) with an explicit external Environment boundary; cross-layer Control, Trust and Adaptation planes; Processing Episodes; recursive delegation semantics; canonical trace objects; authority/uncertainty/effect invariants; and testable conformance classes. Whether this combination is sufficiently distinct and useful for standardization is a matter for expert and committee review.

DIN/ISO fit

DIN identifies AI standardization within NIA and the joint DIN/DKE AI committee NA 043-01-42 GA. ISO/IEC JTC 1/SC 42 maintains core AI standards including 22989 and 23053. The DIN committee should determine whether the proposal belongs as a national DIN project, a contribution to an existing international work item, or a new ISO/IEC proposal.

Research gaps before international proposal

- Licensed full-text crosswalk to relevant ISO/IEC standards.
- Search of IEC, ETSI, CEN/CENELEC and IEEE projects beyond the high-priority set.
- Academic systematic literature review for reference architectures of autonomous/autentic systems.
- Patent/prior-art search only if intellectual-property protection is contemplated; standardization novelty is not patent novelty.

Sources reviewed

DIN Normungsantrag: <https://www.din.de/de/mitwirken/normungsantrag>

DIN AI standardization / NIA: <https://www.din.de/de/mitwirken/normenausschuesse/nia/normungsarbeit-zu-kuenstlicher-intelligenz>

DIN 820 rules overview: <https://www.din.de/de/regeln-der-normung-187188>

ISO/IEC JTC 1/SC 42 catalogue: <https://www.iso.org/committee/6794475/x/catalogue/>

ISO/IEC 22989:2022: <https://www.iso.org/standard/74296.html>

ISO/IEC 23053:2022: <https://www.iso.org/standard/74438.html>

ISO/IEC 5338:2023: <https://www.iso.org/standard/81118.html>

ISO/IEC 42001:2023: <https://www.iso.org/standard/42001>

ISO/IEC 23894:2023: <https://www.iso.org/standard/77304.html>

ISO/IEC/IEEE 42010:2022: <https://www.iso.org/standard/74393.html>

NIST AI RMF: <https://www.nist.gov/itl/ai-risk-management-framework>

NIST 4D/RCS 2002: <https://www.nist.gov/publications/4drcs-reference-model-architecture-intelligent-unmanned-ground-vehicles>

IEEE P3225.05: <https://standards.ieee.org/ieee/3225.05/12760/>

IJCAI BDI survey: <https://www.ijcai.org/proceedings/2020/684>